MedSafe AI: Intelligent Medication Safety & Health Risk Awareness System

LAKSHMI NARAIN COLLEGE OF TECHNOLOGY & EXCELLENCE BHOPAL
Artificial Intelligence and Machine Learning

Submitted To:
SKILL WALLET

Submitted By:

Vaibhav Jaiswal

Yash Vishwari

Yash Kushwaha

Gajendra Kumar Singh

Academic Year: 2026

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Abstract

MedSafe AI is an intelligent healthcare assistance platform designed to enhance medication safety awareness,
symptom understanding, and early health risk identification. The system integrates modern artificial intelligence
technologies including Optical Character Recognition (OCR), fuzzy string matching, curated pharmaceutical
databases, rule-based reasoning, and large language models to help users understand medicines and prescriptions.

The platform allows users to upload prescription images, extract medicine names, identify active salts, check
potential drug–drug interactions, and receive understandable explanations regarding possible risks. The goal of
the system is not to replace professional medical consultation but to improve awareness and prevent medication
misuse through accessible digital tools.

MedSafe AI demonstrates how computer vision, natural language processing, and knowledge-based reasoning
can work together to create a practical healthcare assistance system for patients, caregivers, and health-tech
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Chapter 1: Introduction

1.1 Motivation

Medication errors and lack of awareness about drug interactions remain major challenges in healthcare.
Patients often take multiple medications simultaneously without fully understanding possible interactions,
side effects, or dosage instructions. In many cases prescriptions are handwritten and difficult to read,
which creates confusion for patients and caregivers.

Advancements in artificial intelligence and digital health technologies provide an opportunity to reduce
these problems. AI systems can analyze prescription data, recognize medicine names, and provide educational
information to users in an accessible manner.

MedSafe AI is motivated by the need to create a user-friendly system that helps individuals understand
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1.2 Scope

The scope of MedSafe AI focuses on medication awareness and preventive healthcare education.
The system allows users to upload prescription images and automatically extract medicine names
using OCR technology. These medicines are then analyzed against pharmaceutical datasets to detect
possible interactions and safety concerns.

The platform also provides symptom-based health guidance using AI-powered explanations.
Although it does not provide medical diagnosis, it helps users understand possible causes of symptoms and suggests when professional medical consultation may be necessary.

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1.3 Objectives

The main objective of MedSafe AI is to develop a digital platform that assists users in safely understanding medication usage. The system aims to combine multiple AI technologies including
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Another objective is to provide clear and understandable explanations for complex medical information. Many patients struggle to interpret technical medical terminology; therefore, the system generates simplified explanations that improve comprehension and awareness.

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1.4 Applications

MedSafe AI can be applied in multiple healthcare scenarios including medication awareness, pharmacy education, healthcare research, and patient support systems. Individuals can use the platform to verify medicine names, check interactions, and understand prescription instructions.

Healthcare students and researchers may also use the system to study how artificial intelligence can assist in healthcare safety monitoring. The platform can serve as a foundation for future AI-driven digital health applications.

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Chapter 2: Literature Survey

2.1 Digital Health Systems

Digital healthcare systems have transformed the way medical information is managed and distributed. Electronic health records, telemedicine platforms, and health monitoring applications have significantly improved healthcare accessibility.

Despite these advancements, many systems remain complex and are designed primarily for medical professionals rather than patients. There is still a need for simple, accessible tools that help ordinary users understand medication safety.

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2.2 OCR in Healthcare

Optical Character Recognition (OCR) technology converts printed or handwritten text into machine-readable digital text. In healthcare applications OCR is commonly used to digitize medical records, prescriptions, and clinical documents.

Using OCR for prescription analysis enables automated extraction of medicine names from images, reducing manual data entry and improving information accessibility.

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2.3 Drug Interaction Detection

Drug-drug interaction detection systems analyze combinations of medicines to determine whether they may produce harmful effects. Pharmaceutical databases contain structured information about active ingredients, contraindications, and known interactions.

By comparing extracted medicine names with such datasets, systems like MedSafe AI can identify potentially risky combinations and inform users about safety concerns.

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Chapter 3: Problem Statement

Medication misuse and misunderstanding of prescriptions remain common issues worldwide.
Patients often rely on handwritten prescriptions that may be difficult to interpret.
When multiple medicines are taken simultaneously the risk of drug interactions increases.

Existing healthcare systems typically require professional medical infrastructure and are not always accessible to the general public. As a result many individuals lack tools that help them safely understand medications and possible risks.

MedSafe AI addresses this problem by providing an AI-powered platform that extracts medicine information from prescriptions, checks interactions, and explains possible health risks in a simple and understandable manner.

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Chapter 4: Hardware & Software Requirements

4.1 Software Requirements

Operating System: Windows, Linux, or macOS
Programming Language: Python
Framework: Streamlit
Libraries: OpenCV, pytesseract, pandas, numpy
Database: Pharmaceutical datasets stored in CSV or SQL databases

These tools provide the necessary infrastructure for implementing OCR processing, data analysis, and AI-powered explanations.

Operating System: Windows, Linux, or macOS
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4.2 Hardware Requirements

Processor: Intel i3 or equivalent or higher
RAM: Minimum 4GB (8GB recommended)
Storage: At least 2GB available disk space
Internet Connection: Required for downloading AI models and accessing APIs

The system is designed to operate efficiently on standard consumer hardware, making it accessible for educational projects and research demonstrations.

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Chapter 5: Methodology

The development of MedSafe AI follows an iterative and modular methodology. The first phase involves requirement analysis where key system features are identified. These include prescription OCR extraction, medicine recognition, interaction analysis, symptom interpretation, and user interface design.

After defining requirements, the system architecture is designed to separate different modules including image processing, medicine matching, risk analysis, and explanation generation.

Each module is implemented independently and later integrated into a unified workflow that processes user input and generates structured outputs.

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Chapter 6: System Design Framework

The MedSafe AI architecture consists of several interconnected modules.
The user interacts with the system through a web interface built using Streamlit.
When a prescription image is uploaded, the OCR module extracts text and identifies potential medicine names.

The extracted names are compared against a curated pharmaceutical database using fuzzy matching algorithms. This step ensures that minor spelling errors introduced during OCR processing do not prevent medicine recognition.

After identifying medicines, the system analyzes possible interactions and generates AI-based explanations describing potential risks and safety advice.

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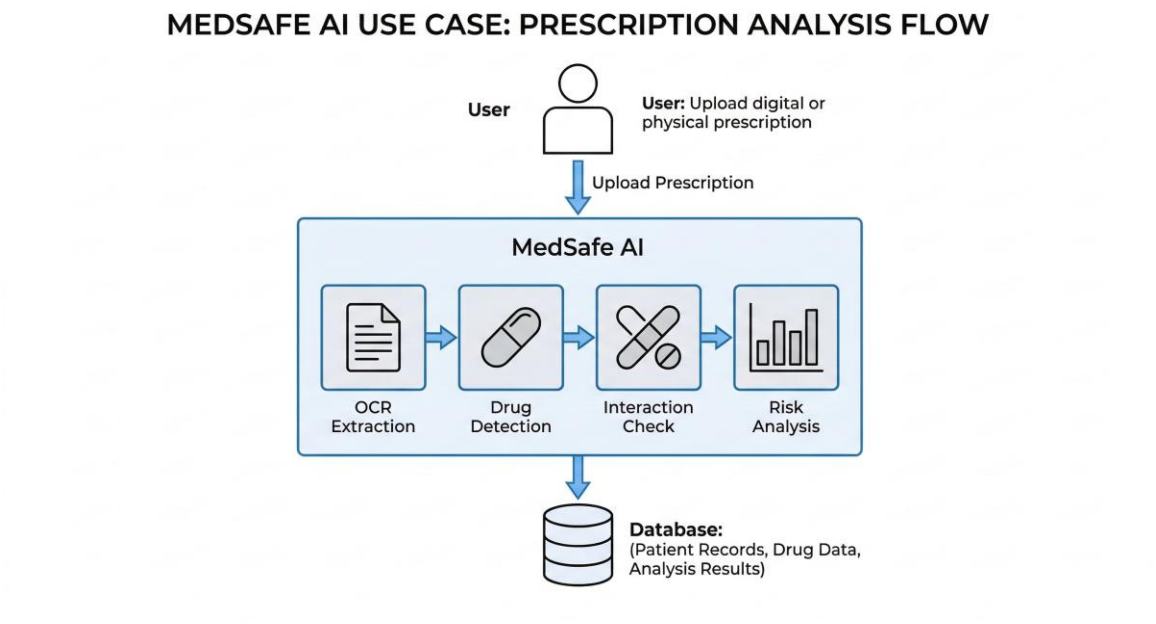
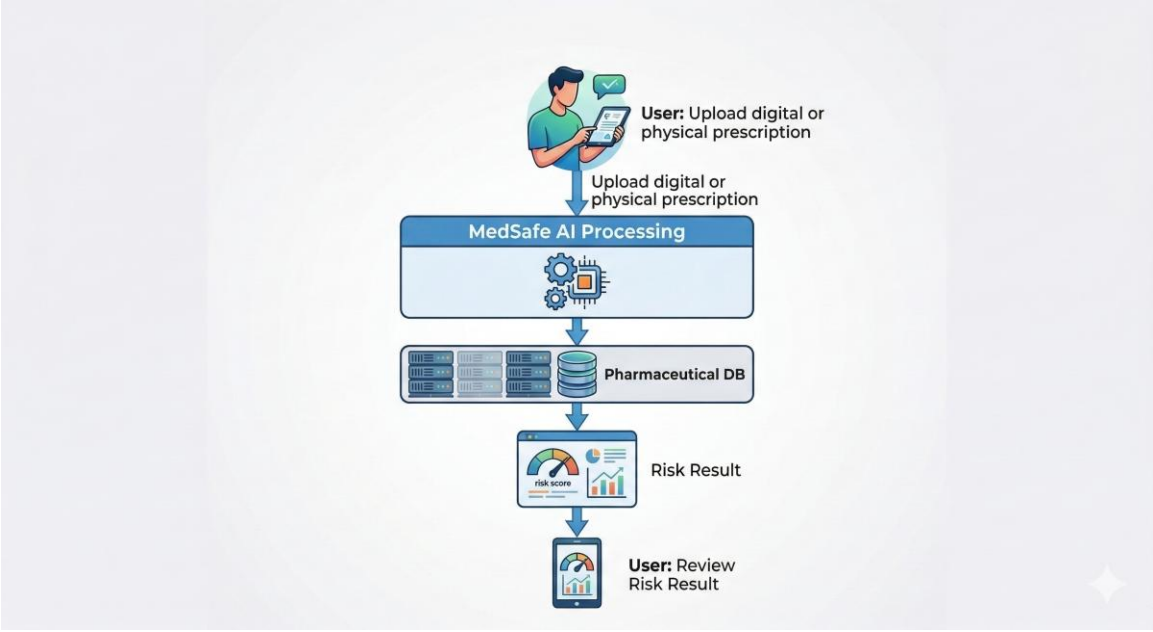
USE CASE DIAGRAM



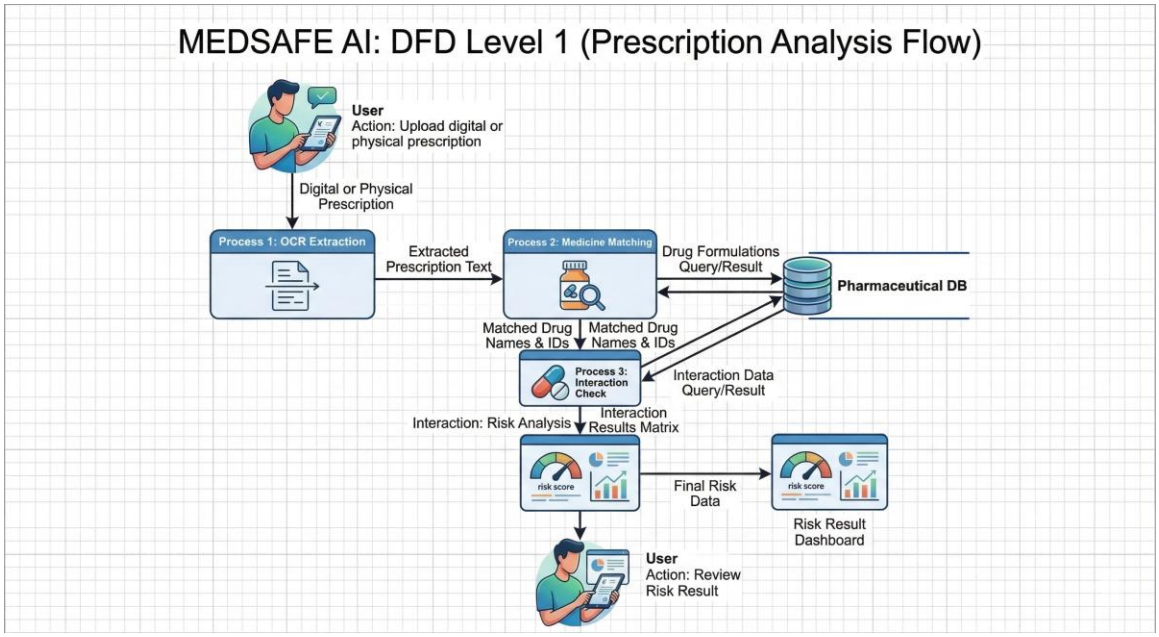
1.1 Use case diagram

DFD LEVEL 0 DIAGRAM

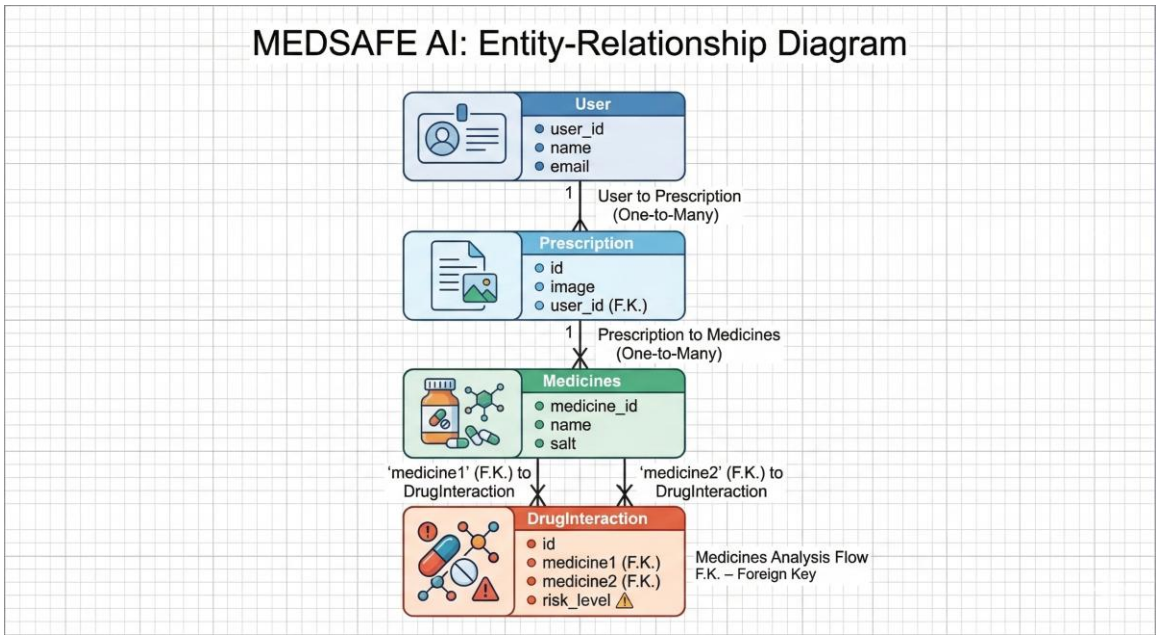


1.2 DFD Level 0

DFD LEVEL 1



ER DIAGRAM



Chapter 7: Implementation

The implementation of MedSafe AI is performed using Python programming language. Streamlit is used for building the web interface, while OpenCV assists in image preprocessing tasks such as resizing, filtering, and contrast enhancement.

Tesseract OCR extracts text from prescription images and converts it into machine-readable format. The extracted text is cleaned and processed before being compared with the medicine database.

Fuzzy matching algorithms help identify medicine names even when OCR introduces small recognition errors. Once medicines are detected, interaction analysis and AI explanation modules generate informative outputs for the user.

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Chapter 8: Testing

Testing is performed to ensure that each module of the MedSafe AI system operates correctly and produces reliable outputs. OCR extraction accuracy is evaluated using multiple prescription samples.

Interaction detection is verified by comparing system outputs with known drug interaction datasets. User interface testing ensures that the platform remains easy to use and responsive across devices and browsers.

Both functional testing and usability testing are performed to validate the overall performance of the system.

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Chapter 9: Conclusion & Future Scope

MedSafe AI demonstrates the potential of artificial intelligence in improving medication safety awareness and healthcare education. By integrating OCR, fuzzy matching algorithms, pharmaceutical databases, and AI explanation models, the system provides users with valuable insights into prescription information and possible drug interactions.

The project highlights how AI technologies can assist individuals in making more informed healthcare decisions and reducing the risk of medication misuse.

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Future Scope

Future improvements may include integration with electronic health record systems, mobile application development, multilingual support, and real-time medical knowledge updates. Machine learning models could also be trained on larger prescription datasets to improve medicine recognition accuracy.

Additional modules such as dosage monitoring, patient health tracking, and AI-powered medical chat assistants could further expand the capabilities of the MedSafe AI platform.

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